

Appendix 3: Final Project & Lab Report information

Background Information

Your Independent Project should be designed to be carried out in the four laboratory periods indicated on the course syllabus. You can work with your current lab partner or a different lab partner. It may be possible to work individually, depending on other groups' selections and availability of equipment. Your Independent Project must be carried out in the teaching lab. Your proposed project can be an extension of a project done previously in the teaching lab or could be a project we have not done previously, if practical to do so in the time available with the equipment available.

Timeline

To clarify the issues involved in pursuing your proposed Independent Project it is imperative to adhere to the following timeline for the development of your ideas regarding your Independent Project:

- **Friday March 27, 9am:** Submit a top 3 rank-ordered list of the experimental systems you'd like to work on with your partner for your final project. For each system you propose, estimate the number of animals/subjects you will require for the experiment you have in mind. If you submit your list by Friday 5pm, you will receive a confirmation of your experimental system on Saturday March 28.
- **Tuesday March 31, 11:59pm:** Draft and submit a brief proposal of your Independent Project (see below).
- **Wed-Thu April 1-2:** Final project conferences: during the second session of this module, you will have a brief meeting with the instructors to discuss the practicalities of your proposed work.
- **Week of Monday April 13:** Start work on your Independent Project.

Recall that your lab report, due May 9 at 7pm, will be based entirely on your Independent Project. The "instructor's discretion" extension for semester-end work is available to all students (i.e., the portal on Canvas will remain open for 24 hours beyond the May 9, 7pm deadline).

Proposal

The proposal you will submit by March 31 should be about two pages, with about 1 page each for an introduction that reflects a literature search that drives your research question, and 1 page that outlines your experimental methods - both practical and analytical. These sections will (hopefully) form the basis of the Introduction and Methods sections of your final report.

You may request reagents such as those found in the *crawdad Pharmacopia*. Include a part number from a vendor such as Sigma or ThermoFisher in your request. We have an easier time getting things like neuromodulators than things like neurotoxins, and we are not permitted to allow the use of any controlled substances.

If you are working with a partner, you may write the methods section collaboratively, but you should write your intro/lit review independently. Sometimes partners propose to share an experimental prep that will allow them to answer different questions, while other times partners will pursue a single experimental aim. If you are working with a partner, you should share your proposal with your partner once you have both submitted on canvas.

The proposal will be evaluated on the following criteria:

- clarity & significance* of the scientific question;
- use & interpretation of appropriate prior literature;
- experimental design (methodological soundness & rigor);
- the feasibility of the proposed experiment;
- clarity and organization of scientific writing.

* note: *It is not required that your experiment be entirely novel (replicating existing work is permitted).*

Project Conference

On Wednesday and Thursday, April 1 and 2, we will hold conferences in lab to review your proposals. Here you will receive feedback on your experimental design, practicalities of executing the project you proposed, and your written proposal itself.

Final Projects

Projects will be completed during the final 4 lab sessions of the semester. Access to the lab during other lab sections may be possible (generally sought by students performing EEG projects). Access to the lab on the Fridays of these two weeks may also be possible for extenuating circumstances, but you should plan an experiment that can be completed from start to finish in 4 class periods.

You will be graded on your execution of the final project, accounting for both effort and success.

Lab Reports

The goal of your lab report is to present your results clearly and concisely, with appropriate introductory material to motivate the measurements you made and a discussion of the meaning, strengths and weaknesses of your data. **Lab reports are written independently from lab partners.**

- Format: Similar to a Brief Communication in the Journal of Neuroscience.
- Components: Title, Abstract, Introduction, Methods, Results, Discussion, Bibliography
- Length: target of 2,000 words, excluding figure legends and bibliography (4-5 pages of text). Budget approximately one page each for Introduction, Methods, Results and Discussion.
- Abstract should clearly but succinctly convey the reason for making the measurements, what measurements were made, the results obtained, and the conclusions that were made based on those data.
- Introduction should provide a brief discussion of the rationale for making the measurements presented with appropriate citations of relevant primary literature.
- Methods should include all technical and procedural details needed to evaluate the adequacy of the methods used and details needed to reproduce the measurements.

- Results should convey the data that were gathered, including both representative raw data and results of any analyses of raw data, with brief statements of the meaning to be derived from each item of data analysis presented. The results section should be *written*, as opposed to being composed entirely of figures with legends.
- Discussion should give a more thorough treatment of the conclusions to be drawn from the data presented, their relationship to previous relevant results from the literature and very brief suggestions for further work needed to amplify and improve the data presented.
- The bibliography should contain the full citations to articles cited in the text. Make sure that all references in the bibliography are cited in the text and that all citations in the text are represented in the bibliography.
- References should be formatted as specified for publication in the Journal of Neuroscience (https://www.jneurosci.org/content/information-authors#preparing_a_manuscript).
- We encourage you to use a citation manager such as Zotero to manage your papers and bibliography. For more info, see <https://libguides.princeton.edu/zotero>.
- Your lab report should be submitted as a pdf file on Canvas.
- A folder of graded sample Lab Reports is available under Course Materials on Canvas, and a rubric is provided on the assignment page.